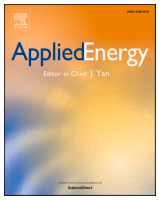
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A two-stage restoration model for power distribution systems with integrated satellite-terrestrial communication networks☆

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HIGHLIGHTS

- A two-stage restoration framework for power distribution systems (PDSs) with an integrated satellite-terrestrial communication network (STCN) is proposed.
- A multi-time period operation simulation model for LEO satellite communication network is established.
- A restoration model is developed that considers the operation constraints of PDSs and STCNs.

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ABSTRACT

With the development of the satellite internet, the low Earth orbit (LEO)-based satellite communication network (SCN) is gradually applied to power grids to meet the emergency communication requirements. By the advantages of SCN, this paper proposes a two-stage restoration framework for power distribution systems (PDSs) with an integrated satellite-terrestrial communication network (STCN). In Stage 1, a multi-time period operation simulation model for LEO satellite constellation is established to derive the available satellite communication links during restoration. Then, a MILP restoration model is developed that considers the operation constraints of PDSs and STCNs in Stage 2 to obtain the optimal PDS restoration strategy under cyber-physical interaction. The STCN includes an SCN and a terrestrial communication network (TCN), which are coordinated to transmit the operation information of PDSs. Finally, the effectiveness of the optimal restoration strategy is verified by constructing a constellation with 1000 LEO satellites for the PDSs in a practical area. Also, the influences of LEO satellite operation parameters are analyzed in the case studies.

1. Introduction

The advancement of information and communication technologies proceeds the development of cyber-physical power systems (CPPSs). These technologies also improve the controllability and operational efficiency of power distribution systems (PDSs) with renewable energy sources. However, the dependence on communications of PDS operations communications introduces various risks of cyber-physical cascading failures [1]. The conventional terrestrial communication network (TCN), represented by the optical fiber network, is vulnerable to extreme natural disasters [2]. For example, the ice disaster and earthquake can destroy power lines and optic fibers of the CCPS simultaneously and caused severe cascading failures in power grids.

To enhance the resilience of CPPSs and leverage controllable devices for quick restoration after disasters, cyber-physical collaborative restoration has emerged as a critical research focus [3]. Some researchers explored the assessment methods of CPPS resilience through graph theory [4], complex network theory [5], and cyber-physical interaction models [1], providing valuable insights for resilience enhancement of CPSSs. Furthermore, practical restoration strategies were discussed, such as topology reconfiguration [6,7], repair crew dispatch [8,9] and routing scheduling of communication links [10,11]. In the above studies, cyber restoration in power systems is mainly based on the wired communication networks. With the development of wireless communication technologies, hybrid wireless-aided restoration approaches for power grids are studied. In [12], 5G-based stations are utilized to rapidly

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cover disconnected feeder terminal units (FTUs). In [13], vehicles for emergency communication vehicles are dispatched to rebuild the communication networks. In [14], unmanned aerial vehicles (UAVs) are scheduled to provide wireless communication services after disasters under an intelligent offloading scheme. These works provide fundamental frameworks for cyber-physical coordinated restoration of PDSs.

In recent years, the low Earth orbit (LEO)-based satellite communication network (SCN) has been widely developed and used for information transmission [15]. As a typical LEO-based satellite internet, the Starlink constellation built by SpaceX has been used to construct stable communication links for power grid operations, especially in isolated places [16]. In [17], the researchers proposed a resilience enhancement strategy for CPSSs based on the SCN. Also, the LEO-based SCN is applied to obtain the operation data that aid in performing predictive control in remote microgrids [18] and inspect power line status under extreme disasters [19]. The above studies demonstrate the application potential of SCN in communication services and the coordination of SCN and TCN would provide more flexibility for information interaction in PDSs. Compared with traditional wireless methods, LEO satellite communication has greater resistance and stronger resilience against disasters. Different from the existing studies focusing on the PDS operations with the assistance of TCN, an innovative communication framework and the PDS restoration strategy considering the advantages of SCN need to be further studied. This is also the original intention of this paper and our contributions are summarized as:

1) A two-stage restoration framework for PDSs with integrated satellite-terrestrial communication networks (STCNs) is proposed, which is totally different from the restoration frameworks in [6]– [13] only considering the application of TCN. The operation simulation for LEO satellites is performed in Stage 1 and an optimal restoration strategy for PDSs is obtained in Stage 2 under the supports of STCNs.

2) A multi-time period operation simulation model for LEO satellites is established to derive the connection states of satellites in Stage 1. The satellite connection states determine the available satellite communication links, which are the inputs to support the PDS restoration in Stage 2. In the simulation model, the influences of LEO satellite operation parameters including the signal loss, time delay of communication, and elevation angle on the satellite connection states are considered.

3) An optimization model considering the cyber-physical interaction process is developed in Stage 2 to derive the optimal PDS restoration strategy. The effectiveness of the restoration strategy is verified by constructing a constellation with 1000 LEO satellites for the PDSs in a practical area.

The remaining parts of this paper are organized as follows. Section II introduces the proposed restoration framework for PDSs with integrated STCNs. Section III establishes a two-stage model to derive the optimal PDS restoration strategy. Section IV uses three cases to demonstrate the effectiveness of the proposed method, and Section V concludes the work.

2. Proposed restoration framework for PDSs

In this section, the structure of PDSs with integrated STCNs is introduced. Then, a two-stage restoration framework for the PDSs is proposed, in which satellite communication links are used to improve the efficiency of PDS restoration.

2.1. PDSs with integrated STCNs

The structure of PDSs with integrated STCNs is presented in Fig. 1 and the details are introduced as follows:

The STCN consists of a terrestrial communication network (TCN) and a satellite communication network (SCN). The TCN includes sensors, switches, information centres and terrestrial communication links. Each switch connects to multiple sensors and each sensor has a “one-to-one” correspondence with a PDS bus as shown in Fig. 1 [20]. Switches are utilized to transmit the data between sensors and the information centers through the terrestrial communication links [21]. The SCN is constructed by the LEO satellites [22], which transmit the data through the satellite communication links as shown in Fig. 1 [23]. The antennas are the important devices to establish the satellite communication links, which are assigned at the nodes of information centres and sensors.

By STCNs, the operation states and control orders of the PDSs can be transmitted through the terrestrial and satellite communication links. If the communication links between the sensor allocated at the PDS bus and the information centre are all accessible, we define that the buses as well as the devices connecting to the bus are *observable*. When the related terrestrial communication links are damaged, those buses or devices as shown in Fig. 1 would become unobservable until the

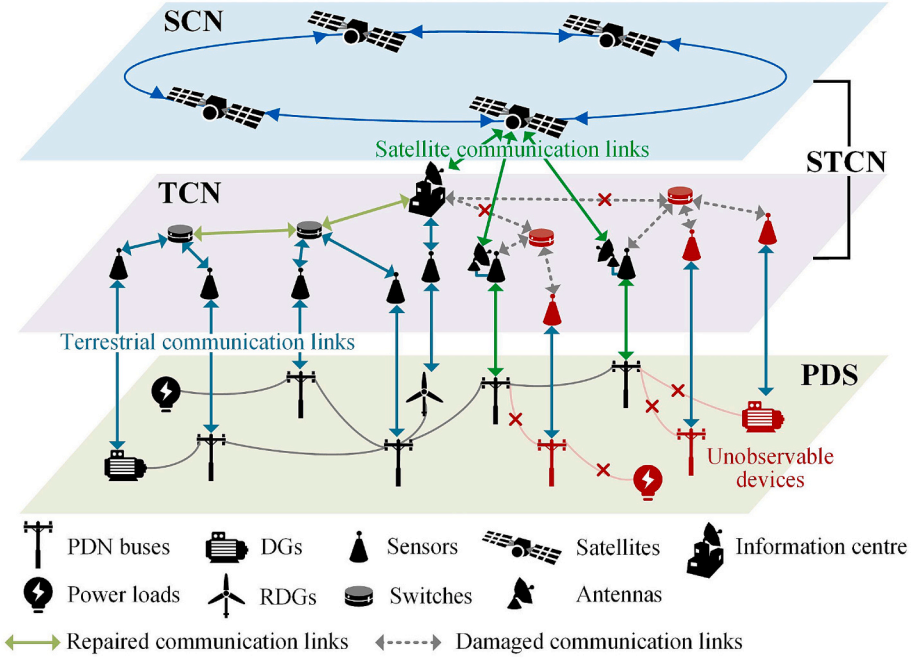


Fig. 1. The structure of PDSs with integrated STCNs.

damaged terrestrial communication links are repaired by the maintenance crew or replaced by the available satellite communication links. In practical operation, the observability of buses influences the controllability of circuit breakers, distributed generators (DGs), and loads in PDSs. As shown in Fig. 2, the circuit breaker in a feeder is controllable when the two buses connecting to the breaker are observable. Also, the states of DGs and loads are controllable if the corresponding buses are observable. During restoration, the PDS operator can close the controllable circuit breakers to form DG-centered radial microgrids so that the crucial loads can be restored. In the restoration process, the STCN is utilized to transmit measurement and control information in PDSs, which directly affects the observability and controllability of buses and various devices in PDSs after disasters.

2.2. Restoration Framework for PDSs with Integrated STCNs

In this part, we propose a two-stage restoration framework for PDSs with integrated STCNs as shown in Fig. 3. In Stage 1, the available satellite communication links are derived to support the information interaction in PDS restoration. In Stage 2, an optimization model for PDSs with integrated STCNs to obtain the optimal restoration strategy. The two-stage restoration framework is described as follows:

Stage 1: An operation simulation model for LEO satellites is developed to determine the positions of LEO satellites during the restoration time periods. The LEO satellite operation parameters are the input of the simulation model. Then, the available satellite communication links are obtained according to the real-time satellite connection statuses represented by a group of binary variables, then inputted into stage 2.

Stage 2: A MILP model is built considering the constraints of the PDS and STCN operations. The objective of the model is to maximize the restored loads in PDSs. The available satellite communication links derived in Stage 1, TCN restoration process, topology reformulation, power flow, and cyber-physical interaction process between the PDSs and STCNs are all represented by various constraints in the model.

2.3. Model Assumptions

The assumptions of this paper are given as follows:

1. The PDS operator can adjust the restored loads on the observable buses to satisfy the power flow constraints.
2. The terrestrial communication links can be repaired by the maintenance crew during restoration and repair of the damaged power lines is not considered. This focuses the model on the rapid initial cyber-reconfiguration phase.
3. Due to the characteristics of satellite communication, only buses equipped with satellite antennas can connect to the LEO satellites. Satellite communication links also have limits in transmission capacity.

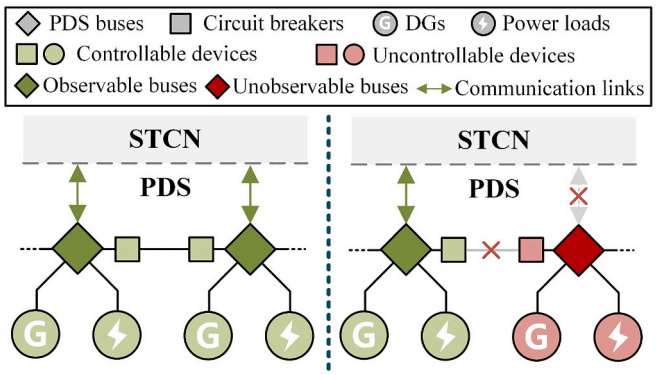


Fig. 2. Relationship of controllable devices and the observable buses.

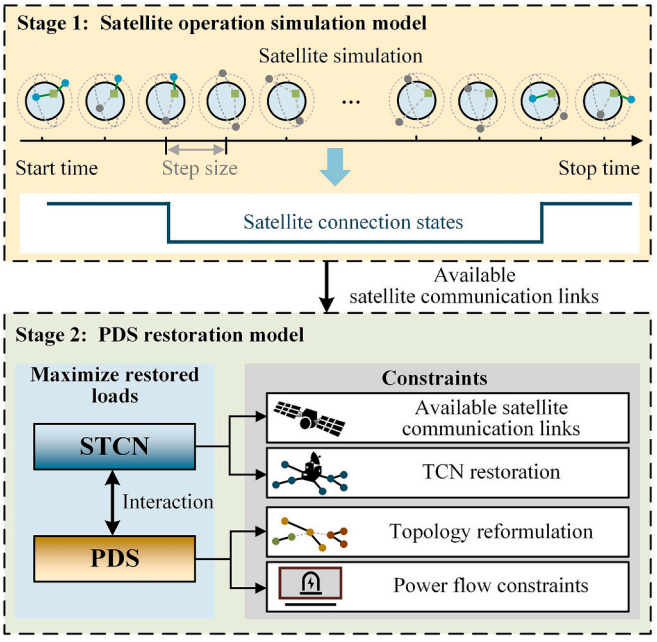


Fig. 3. The two-stage restoration framework for PDSs with integrtaed STCNs.

3. Model formulation

In this section, the satellite operation simulation model in Stage 1 and the PDS restoration model in Stage 2 are developed.

3.1. Stage 1: Satellite Operation Simulation Model

In each simulation time period t_k, the connection state of LEO satellite a is represented by a binary variable A_k^{S}:

$$d_{\max} = \min\left[d_{\max}^{\mathrm{L}}, d_{\max}^{\mathrm{D}}, d_{\max}^{\mathrm{A}}\right] \tag{1}$$

$$A_k^{\mathrm{S}} = \begin{cases} 1, d_{ax,k} \le d_{\max}, \exists a \in S^{\mathrm{C}} \\ 0, \text{otherwise} \end{cases}, \quad \forall t_k, \forall x \in S^{\mathrm{G}} \tag{2}$$

where $d_{\max}$ is the distance required to build available satellite communication links; S^{C} is the set of satellites; S^{G} is the set of ground communication devices including the antennas and the information centres; $t_k = t_0 + k\delta$ is the kth simulation time period and t_0 is the start time; δ is the simulation step size; $d_{ax,k}$ is the distance between the LEO satellite a and the ground communication device x in t_k; $d_{\max}^{\mathrm{L}}$, $d_{\max}^{\mathrm{D}}$ and $d_{\max}^{\mathrm{A}}$ are the maximum connection distance under the requirement of signal loss, time delay and elevation angle, respectively.

To derive the satellite connection states in (2), four parameters $d_{ax,k}$, $d_{\max}^{\mathrm{L}}$, $d_{\max}^{\mathrm{D}}$ and $d_{\max}^{\mathrm{A}}$ need to be calculated:

1. The distance $d_{ax,k}$ between the LEO satellite a and the ground communication device x is calculated using the orbit propagator model [24], which is stated as

$$d_{ax,k} = O(\boldsymbol{x}_a, \boldsymbol{l}_x, t_k), \quad \forall t_k, a \in S^{\mathrm{C}}, x \in S^{\mathrm{G}} \tag{3}$$

where $\boldsymbol{x}_a$ is the orbital parameters of satellite a; $\boldsymbol{l}_x$ includes the latitude and longitude of the ground communication device x; $O(\cdot)$ is a function to derive the distance referring to [25].

2. Signal loss and time delay in the space also constrain the maximum transmission distance of satellite communication. Considering the required signal intensity and latency of STCNs, the maximum connection distance $d_{\max}^{\mathrm{L}}$ and $d_{\max}^{\mathrm{D}}$ can be calculated as presented in [26].

3. The elevation angle ε affects the availability of satellite communication links. The minimal elevation angle for satellite communication

in the Starlink LEO satellite constellation is 40° [27]. Given the minimal elevation angle $\varepsilon_{\min}$, the maximum connection distance $d^{\mathrm{A}}_{\max}$ is stated as:

$$d^{\mathrm{A}}_{\max} = R\left[\sqrt{(H+R)^2 \Big/ R^2 - \cos^2\varepsilon_{\min}} - \sin\varepsilon_{\min}\right] \tag{4}$$

where H is the orbital altitude of satellite as shown in Fig. 4; R = 6356.91 km is the radius of the earth.

By $d_{\max}$ obtained in (1), Eq. (2) indicates that the satellite communication links are available to support the PDS restoration in the kth simulation time period if and only if there exists a LEO satellite a whose position satisfies the distance constraints. In Fig. 5, the satellite communication link related to satellite a is available, while those related to satellites b and c are unavailable. The reason is that the distance between the satellite b/c and the ground communication device x is greater than the distance required $d_{\max}$. Furthermore, it is worth noting that each time period τ in Stage 2 contains N simulation step sizes of Stage 1. We define the availability rate of satellite communication in each time period τ and the satellite communication links can be used for PDS restoration when the availability rate is greater than a limit $\eta_{\min}$. The relationship between the availability rate and the satellite connection state variable is shown in Fig. 6 and stated as:

$$\eta^{\mathrm{S}}_m = \frac{\sum_{i=mN+1}^{(m+1)N} A^{\mathrm{S}}_i}{N}, \quad \forall m$$
$$C^{\mathrm{S}}_m = \begin{cases} 1, \eta^{\mathrm{S}}_m \geq \eta_{\min} \\ 0, \eta^{\mathrm{S}}_m < \eta_{\min} \end{cases}, \quad \forall m \tag{5}$$

where η^{S}_m is the availability rate of satellite communication in mth time period; C^{S}_m represents the availability of the satellite communication link and $C^{\mathrm{S}}_m = 1$ indicates that the satellite communication link is available in mth time period.

3.2. Stage 2: PDS Restoration Model

A graph $\boldsymbol{G}^{\mathrm{P}} = (N^{\mathrm{P}}, L^{\mathrm{P}})$ is used to depict the PDSs, where N^{P} is the set of buses and L^{P} is the set of lines. A graph $\boldsymbol{G}^{\mathrm{C}} = (N^{\mathrm{C}}, L^{\mathrm{C}})$ is used to represent the topology of the STCN, where N^{C} is the set of nodes and L^{P} is the set of communication links. The PDS restoration model aims to maximize the recovered loads and the objective function is stated as:

$$\max \sum_{t=1}^{T}\sum_{i\in N^{\mathrm{P}}} w_i P^{\mathrm{L}}_{i,t}\tau \tag{6}$$

where $w_i \in [0,1]$ is the weight coefficient representing the importance of the loads on bus i; $P^{\mathrm{L}}_{i,t}$ is the restored loads at bus i in the tth time period; τ is the restoration time step.

The constraints of the PDS restoration model include:

1. Constraints for PDS

1) *Topology reformulation constraints:* The DGs are used to restore the cutoff loads by black start, and multiple islanded microgrids would be formed through topology reformulation. In this paper, the Energization Agent (EA) method [28] is used to describe the topology reformulation constrains for the islanded microgrid operation with the black-start DGs. For two black-start DGs, the topology reformulation process by the EA method is shown in Fig. 7. In Fig. 7(a), the EA starts moving from the black-start DGs and progressively connects the adjacent buses to form islanded microgrids. The restoration path can be mathematically represented with a sparse $|N^{\mathrm{P}}| \times |N^{\mathrm{P}}|$ route table of binary variables, as shown in Figs. 7(b) and (c). By the EA method, the constraints are formulated as:

$$x^{\mathrm{P}}_{ii} = \begin{cases} 1, & \forall i \in N^{\mathrm{P}}_{\mathrm{G}} \\ 0, & \forall i \notin N^{\mathrm{P}}_{\mathrm{G}} \end{cases} \tag{7}$$

$$x^{\mathrm{P}}_{ij} = 0, \quad \forall i \notin N^{\mathrm{P}}_{\mathrm{G}}, j \in N^{\mathrm{P}}_{\mathrm{G}} \tag{8}$$

$$x^{\mathrm{P}}_{ij} + x^{\mathrm{P}}_{ji} \leq 1, \quad \forall (i,j) \in L^{\mathrm{P}}, i \neq j \tag{9}$$

$$\sum_{i=1}^{n^{\mathrm{P}}} x^{\mathrm{P}}_{ij} \leq 1, \quad \forall j \in N^{\mathrm{P}} \tag{10}$$

$$\sum_{j=1}^{n^{\mathrm{P}}} x^{\mathrm{P}}_{ij} \leq n^{\mathrm{P}} \sum_{h=1}^{n^{\mathrm{P}}} x^{\mathrm{P}}_{hi}, \quad \forall i \in N^{\mathrm{P}} \tag{11}$$

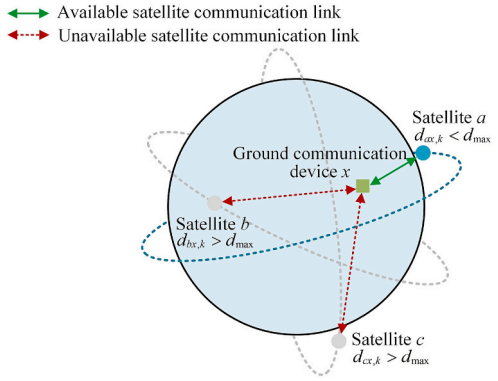


Fig. 5. Available and unavailable satellite communication links.

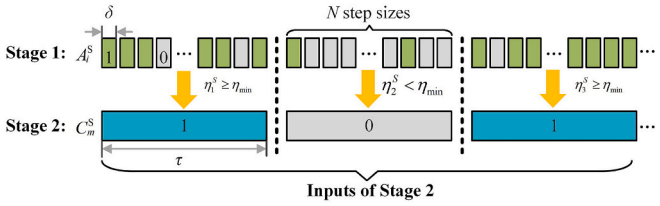


Fig. 6. Relationship between the states of the satellite connection and the availability of the satellite communication link.

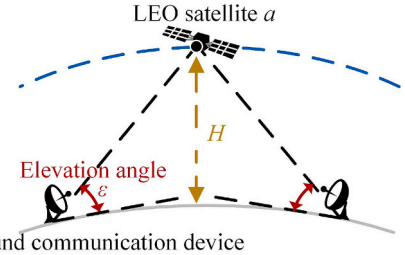


Fig. 4. Elevation angle ε of the ground communication device.

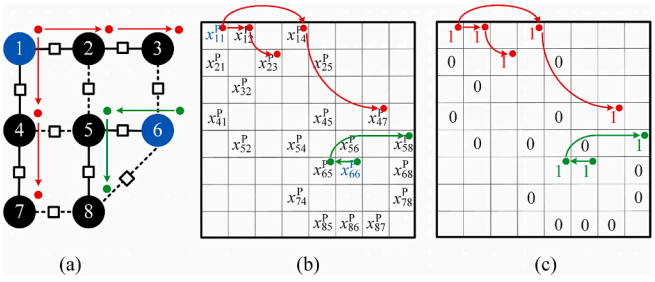


Fig. 7. Application of the EA method for two black-start DGs at buses 1 and 6.

$$x_{ij}^{\mathrm{P}} \le M_{ij}^{\mathrm{P}},\ \ \forall i,j \in N^{\mathrm{P}} \tag{12}$$

where $N_{\mathrm{G}}^{\mathrm{P}}$ is the set of DGs; $n^{\mathrm{P}} = |N^{\mathrm{P}}|$ is the number of the PDS buses; $\boldsymbol{M}^{\mathrm{P}} = \left[M_{ij}^{\mathrm{P}}\right]^{|N^{\mathrm{P}}|\times|N^{\mathrm{P}}|}$ is the adjacency matrix of the power lines, where M_{ij}^{P} is a binary variable and $M_{ij}^{\mathrm{P}} = 1$ indicates the power line between bus i and bus j has been restored.

Constraint (7) indicates that the EA starts moving at the buses with black-start DGs. Constraints (8)–(10) restrict the radial topology of microgrids. Constraint (11) indicates that the restoration of child buses is strictly contingent upon the prior restoration of their parent bus. Constraint (12) restricts that the EA moves through the restorable power lines.

2) *Restoration time constraints:* The restoration of loads at different PDS buses is sequential considering the status of the power lines and electricity supply of the DGs. The restoration time constraints for the PDS buses are stated as follows:

$$t_i^{\mathrm{P}} = \min\left(t_i^{\mathrm{P,C}}, t_i^{\mathrm{S,a}}\right) + T_{ii}^{\mathrm{P}},\ \ \forall i \in N_{\mathrm{G}}^{\mathrm{P}} \tag{13}$$

$$t_{ij}^{\mathrm{R}} = \max\left[t_i^{\mathrm{P}}, \min\left(t_j^{\mathrm{P,C}}, t_j^{\mathrm{S,a}}\right)\right],\ \ \forall (i,j) \in L^{\mathrm{P}} \tag{14}$$

$$-M\left(1 - x_{ij}^{\mathrm{P}}\right) \le t_j^{\mathrm{P}} - t_{ij}^{\mathrm{R}} - T_{ij}^{\mathrm{P}} \le M\left(1 - x_{ij}^{\mathrm{P}}\right), \forall (i,j) \in L^{\mathrm{P}}, j \notin N_{\mathrm{G}}^{\mathrm{P}} \tag{15}$$

$$(T+1) - M\sum_{i=1}^{n^{\mathrm{P}}} x_{ij}^{\mathrm{P}} \le t_j^{\mathrm{P}} \le (T+1) + M\sum_{i=1}^{n^{\mathrm{P}}} x_{ij}^{\mathrm{P}},\ \ \forall j \notin N_{\mathrm{G}}^{\mathrm{P}} \tag{16}$$

$$\sum_{t=1}^{T+1} e_{i,t}^{\mathrm{P}} \le M\sum_{h=1}^{n^{\mathrm{P}}} x_{hi}^{\mathrm{P}},\ \ \forall i \in N^{\mathrm{P}}, \forall t \tag{17}$$

$$T + 1 - \sum_{t=1}^{T+1} e_{i,t}^{\mathrm{P}} = t_i^{\mathrm{P}},\ \ \forall i \in N^{\mathrm{P}}, \forall t \tag{18}$$

$$e_{i,t-1}^{\mathrm{P}} \le e_{i,t}^{\mathrm{P}},\ \ \forall i \in N^{\mathrm{P}}, \forall t \ge 2 \tag{19}$$

$$l_{k,t}^{\mathrm{P}} = e_{i,t}^{\mathrm{P}} e_{j,t}^{\mathrm{P}}\left(x_{ij}^{\mathrm{P}} + x_{ji}^{\mathrm{P}}\right),\ \ \forall k = (i,j) \in L^{\mathrm{P}}, \forall t \tag{20}$$

where t_i^{P} is the restored time of the loads at bus i; T_{ij}^{P} is the required time to close the controllable circuit breaker between buses i and j; T_{ii}^{P} is the self-start time of the black-start DG at bus i; t_{ij}^{R} is the time when bus i is restored as well as bus j is observable; $t_j^{\mathrm{P,C}}$ is the time when the node of TCN connecting to bus i is restored; $t_j^{\mathrm{S,a}}$ is the time of constructing satellite communication links for bus j; M is a large constant; $e_{i,t}^{\mathrm{P}}$ and $l_{k,t}^{\mathrm{P}}$ are the binary variables to represent the restoration statuses of buses and power lines in PDSs, where $e_{i,t}^{\mathrm{P}} = 1$ ($l_{k,t}^{\mathrm{P}} = 1$) indicates the loads at bus i (power line k) are restored at time t.

Constraint (13) restricts the restored time of the loads at the bus with black-start DG. Constraints (14) and (15) indicate that the loads on bus j can be restored when there is a restoration path from bus i to bus j, which means that the circuit breaker between buses i and j is controllable. Constraint (16) represents that the time to restore bus j is set to be $T+1$ if it has not been restored. Constraints (17)–(20) represent the relationship between the restoration time and the restoration statuses of buses and power lines in each time period.

3) *Power flow constraints*: The linearized Dist-flow model [29] is used to constrain the power flow, ensuring that the reformulated microgrids satisfy the secure operation requirements. The power flow constraints are stated as:

$$P_i^{\mathrm{G,min}} \le P_{i,t}^{\mathrm{G}} \le P_i^{\mathrm{G,max}},\ \ \forall i \in N_{\mathrm{G}}^{\mathrm{P}}, \forall t \tag{21}$$

$$Q_i^{\mathrm{G,min}} \le Q_{i,t}^{\mathrm{G}} \le Q_i^{\mathrm{G,max}},\ \ \forall i \in N_{\mathrm{G}}^{\mathrm{P}}, \forall t \tag{22}$$

$$R_i^{\mathrm{G}-}\tau \le P_{i,t}^{\mathrm{G}} - P_{i,t-1}^{\mathrm{G}} \le R_i^{\mathrm{G}+}\tau,\ \ \forall i \in N_{\mathrm{G}}^{\mathrm{P}}, \forall t \ge 2 \tag{23}$$

$$0 \le P_{i,t}^{\mathrm{L}} \le e_{i,t}^{\mathrm{L}} P_i^{\mathrm{L,max}},\ \ \forall i \in N^{\mathrm{P}}, \forall t \tag{24}$$

$$Q_{i,t}^{\mathrm{L}} = \eta P_{i,t}^{\mathrm{L}},\ \ \forall i \in N^{\mathrm{P}}, \forall t \tag{25}$$

$$P_{i,t-1}^{\mathrm{L}} \le P_{i,t}^{\mathrm{L}},\ \ \forall i \in N^{\mathrm{P}}, \forall t \ge 2 \tag{26}$$

$$U^{\min} e_{i,t}^{\mathrm{P}} \le U_{i,t} \le U^{\max} e_{i,t}^{\mathrm{P}},\ \ \forall i \in N^{\mathrm{P}}, \forall t \tag{27}$$

$$U_{i,t} = U_0 e_{i,t}^{\mathrm{P}},\ \ \forall i \in N_{\mathrm{G}}^{\mathrm{P}}, \forall t \tag{28}$$

$$-P_k^{\max} l_{k,t}^{\mathrm{P}} \le P_{k,t} \le P_k^{\max} l_{k,t}^{\mathrm{P}},\ \ \forall k \in L^{\mathrm{P}}, \forall t \tag{29}$$

$$-Q_k^{\max} l_{k,t}^{\mathrm{P}} \le Q_{k,t} \le Q_k^{\max} l_{k,t}^{\mathrm{P}},\ \ \forall k \in L^{\mathrm{P}}, \forall t \tag{30}$$

$$\sum_{k \in L^{\mathrm{P}}} \left(P_{k,t} D_{ki}^{\mathrm{P}}\right) = P_{i,t}^{\mathrm{G}} - P_{i,t}^{\mathrm{L}},\ \ \forall i \in N^{\mathrm{P}}, \forall k \in L^{\mathrm{P}}, \forall t \tag{31}$$

$$\sum_{k \in L^{\mathrm{P}}} \left(Q_{k,t} D_{ki}^{\mathrm{P}}\right) = Q_{i,t}^{\mathrm{G}} - Q_{i,t}^{\mathrm{L}},\ \ \forall i \in N^{\mathrm{P}}, \forall k \in L^{\mathrm{P}}, \forall t \tag{32}$$

$$\begin{aligned} &U_{i,t} - \frac{P_{k,t}R_{ij} + Q_{k,t}X_{ij}}{U_0} - M\left(1 - l_{k,t}^{\mathrm{P}}\right) \le U_{j,t}, \forall k = (i,j) \in L^{\mathrm{P}}, \forall t \\ &U_{i,t} - \frac{P_{k,t}R_{ij} + Q_{k,t}X_{ij}}{U_0} + M\left(1 - l_{k,t}^{\mathrm{P}}\right) \ge U_{j,t}, \forall k = (i,j) \in L^{\mathrm{P}}, \forall t \end{aligned} \tag{33}$$

where $P_{i,t}^{\mathrm{G}}$ and $Q_{i,t}^{\mathrm{G}}$ are the active and reactive power outputs of the black-start DG at bus i in time period t, respectively; $P_i^{\mathrm{G,min}}$ and $P_i^{\mathrm{G,max}}$ are the lower and upper limits of the active power outputs of the DG at bus i, respectively; $Q_i^{\mathrm{G,min}}$ and $Q_i^{\mathrm{G,max}}$ are the lower and upper limits of the reactive power outputs of the DG at bus i, respectively; $R_i^{\mathrm{G}-}$ and $R_i^{\mathrm{G}+}$ are the maximum downward and upward ramp rate of DG at bus i, respectively; $P_{i,t}^{\mathrm{L}}$ and $Q_{i,t}^{\mathrm{L}}$ are the restored active and reactive loads at bus i in time period t, respectively; η is the power factor; $P_i^{\mathrm{L,max}}$is the maximum loads which can be restored at bus i; $U_{i,t}$ is the voltage magnitude of bus i at time period t; $U^{\min}$ and $U^{\max}$ are the lower and upper limits of voltage magnitude, respectively; U_0 is the base value of voltage magnitude; $P_k^{\max}$ and $Q_k^{\max}$ are the active and reactive power capacity of power line k, respectively; D_{ki}^{P} is a variable representing the connection relationship between line k and bus i.

Constraints (21)–(23) limit the power outputs of DGs. Constraints (24)–(26) limit the restored loads. Constraints (27) and (28) limit the voltage of buses. Constraints (29)–(30) limit the power line capacity. Constraints (31)–(33) denote the power and voltage drop relationships by linearized Dist-flow model. Also, it is worth noting that if the power line k connects buses i and j and the direction is from i to j, the variables representing the connection relationship satisfy $D_{ki}^{\mathrm{P}} = 1$ and $D_{kj}^{\mathrm{P}} = -1$.

2. Constraints for STCN

1) *Topology reformulation constraints for TCN:* The EA model is also used to depict the restoration process of TCN and the topology reformulation constraints for TCN are stated as:

$$x_{aa}^{\mathrm{C}} = \begin{cases} 1, & \forall a \in N_{\mathrm{B}}^{\mathrm{C}} \\ 0, & \forall a \notin N_{\mathrm{B}}^{\mathrm{C}} \end{cases} \tag{34}$$

$$x_{ab}^{\mathrm{C}} = 0,\ \ \forall a \notin N_{\mathrm{B}}^{\mathrm{C}}, b \in N_{\mathrm{B}}^{\mathrm{C}} \tag{35}$$

$$x_{ab}^{\mathrm{C}} + x_{ba}^{\mathrm{C}} \le 1,\ \ \forall (a,b) \in L^{\mathrm{C}}, a \ne b \tag{36}$$

$$\sum_{a=1}^{n^{\mathrm{C}}} x_{ab}^{\mathrm{C}} \le 1, \ \ \forall b \in N^{\mathrm{C}} \tag{37}$$

$$\sum_{a=1}^{n^{\mathrm{C}}} x_{ab}^{\mathrm{C}} \le n^{\mathrm{C}} \cdot \sum_{h=1}^{n^{\mathrm{C}}} x_{ha}^{\mathrm{C}}, \ \ \forall i \in N^{\mathrm{C}} \tag{38}$$

$$x_{ab}^{\mathrm{C}} \le M_{ab}^{\mathrm{C}}, \ \ \forall a,b \in N^{\mathrm{C}} \tag{39}$$

where $N_{\mathrm{B}}^{\mathrm{C}}$ is the set of nodes with information centres; $\boldsymbol{M}^{\mathrm{C}} = \left[M_{ab}^{\mathrm{C}}\right]^{|N^{\mathrm{C}}|\times|N^{\mathrm{C}}|}$ is the adjacency matrix of the TCN, where $M_{ab}^{\mathrm{C}} = 1$ indicates the communication link between buses a and b has been restored; x_{ab}^{C} is a binary variable to represent the restoration path and x_{ab}^{C}=1 indicates that the restoration sequence is from node a to node b in TCN.

Constraint (34) indicates that the restoration of TCN starts from the information centres. Constraints (35)–(37) restrict the radial topology of TCN. Constraint (38) indicates that the child nodes can be restored after their parent nodes are restored. Constraint (39) indicates that the restoration of TCN depends on the restored communication links.

2) *Restoration time constraints for TCN:* After disasters, the maintenance crew is dispatched to repair the damaged terrestrial communication links and switches. The restoration time constraints for the ground communication devices are stated as:

$$r_a = 1, \ \ a \in N_{\mathrm{B}}^{\mathrm{C}} \tag{40}$$

$$r_a \ge r_{F_a} + 1, \ \ \forall a \in N^{\mathrm{C}} \tag{41}$$

$$I_{d,a} = \begin{cases} 1, & r_a = d \\ 0, & r_a \ne d \end{cases}, \forall a \in N^{\mathrm{C}}, \forall d \tag{42}$$

$$r_a - M\left(1 - x_{ab}^{\mathrm{C}}\right) \le r_{F_b} \le r_a + M\left(1 - x_{ab}^{\mathrm{C}}\right), \forall (a,b) \in L^{\mathrm{C}} \tag{43}$$

$$T_d^{\mathrm{r}} = \begin{cases} 1, & d = 1 \\ T_{d-1}^{\mathrm{r}} + t_d^{\mathrm{r}}, & d \ge 2 \end{cases} \tag{44}$$

$$T_d^{\mathrm{r}} - M\left(1 - I_{d,a}\right) \le t_a^{\mathrm{C}} \le T_d^{\mathrm{r}} + M\left(1 - I_{d,a}\right), \ \ \forall a \in N^{\mathrm{C}}, \forall d \tag{45}$$

$$\sum_{t=1}^{T+1} e_{a,t}^{\mathrm{C}} \le M \sum_{h=1}^{n^{\mathrm{C}}} x_{ha}^{\mathrm{C}}, \ \ \forall a \in N^{\mathrm{C}}, \forall t \tag{46}$$

$$T + 1 - \sum_{t=1}^{T+1} e_{a,t}^{\mathrm{C}} = t_a^{\mathrm{C}}, \ \ \forall a \in N^{\mathrm{C}}, \forall t \tag{47}$$

$$e_{a,t-1}^{\mathrm{C}} \le e_{a,t}^{\mathrm{C}}, \ \ \forall a \in N^{\mathrm{C}}, \forall t \ge 2 \tag{48}$$

$$l_{g,t}^{\mathrm{C}} = e_{a,t}^{\mathrm{C}} e_{b,t}^{\mathrm{C}} \left(x_{ab}^{\mathrm{C}} + x_{ba}^{\mathrm{C}}\right), \ \ \forall g = (a,b) \in L^{\mathrm{C}}, \forall t \tag{49}$$

where r_a is the repair step for node a; d is the repair step number; F_a is the parent node of node a in the restored topology; $I_{d,a}$ is a binary variable representing the status of node a at the repair step d; T_d^{r} is the total time; t_d^{r} is the required time to complete the repair step d; t_a^{C} is the restoration time of node a; $e_{a,t}^{\mathrm{C}}$ and $l_{g,t}^{\mathrm{C}}$ are binary variables representing the restoration status of nodes and lines in TCN, where $e_{a,t}^{\mathrm{C}}$=1 ($l_{g,t}^{\mathrm{C}}$=1) indicates the node a (line g) is restored at time t.

Constraints (40) and (41) indicate that information centres are restored firstly and other nodes are restored at least one repair step after their parent nodes are restored. Constraint (42) defines a binary variable to show if node a is repaired at step d. Constraint (43) indicates that the relationship between the parent and child nodes are determined by the restoration sequence. Constraint (44) presents the relationship between the total repair time and the time cost in each repair step. Constraint (45) presents the restoration time of a TCN node. Constraints (46)–(49) represent the relationships between the restoration time and the statuses of TCN nodes and lines. It is worth noting that the piecewise linearization method can be employed for (44) to obtain the repair time t_d^{r}.

3) *Data flow constraints in TCN:* The data flow constraints of TCN are similar to the power flow constraints in PDSs. Each data switch dispatches data flow proportional to the number of PDS buses. The information centres receive all data flows. Thus, the data flow constraints are stated as:

$$t_a^{\mathrm{C}} \le t_i^{\mathrm{P,C}} \le T + 1, \ \ \forall i \in N^{\mathrm{P}}, a = N_i^{\mathrm{C}} \tag{50}$$

$$T + 1 - \sum_{t=1}^{T+1} e_{i,t}^{\mathrm{P,C}} = t_i^{\mathrm{P,C}}, \ \ \forall i \in N^{\mathrm{P}} \tag{51}$$

$$f_{a,t}^{\mathrm{I}} = \sum_{i \in N_a^{\mathrm{P}}} \left(e_{i,t}^{\mathrm{P,C}} f_i^{\mathrm{C}}\right), \ \ \forall a \in N^{\mathrm{C}}, \forall t \tag{52}$$

$$f_{a,t}^{\mathrm{O}} = \begin{cases} \sum_{h=1}^{n^{\mathrm{C}}} \left(f_{h,t}^{\mathrm{I}}\right), & a \in N_{\mathrm{B}}^{\mathrm{C}} \\ 0, & a \notin N_{\mathrm{B}}^{\mathrm{C}} \end{cases}, \forall t \tag{53}$$

$$\sum_{g \in L^{\mathrm{C}}} \left(f_{g,t} D_{ga}^{\mathrm{C}}\right) = f_{a,t}^{\mathrm{I}} - f_{a,t}^{\mathrm{O}}, \ \ \forall a \in N^{\mathrm{C}}, \forall g \in L^{\mathrm{C}}, \forall t \tag{54}$$

$$-f_g^{\max} l_{g,t}^{\mathrm{C}} \le f_{g,t} \le f_g^{\max} l_{g,t}^{\mathrm{C}}, \ \ \forall g = (a,b) \in L^{\mathrm{C}}, \forall t \tag{55}$$

where t_a^{C} denotes the time when the switch a is restored; $t_i^{\mathrm{P,C}}$ denotes the time when the bus i is observable; N_i^{C} is the TCN node used to transmit the state information of bus i; $e_{i,t}^{\mathrm{P,C}}$ is a binary variable to represent the observability of bus i and $e_{i,t}^{\mathrm{P,C}} = 1$ indicates that the bus i is observable at time t; f_i^{C} is the required flow rate for information transmission of bus i; N_a^{P} is the set of PDS buses connecting to the switch at TCN node a; $f_{a,t}^{\mathrm{I}}$ and $f_{a,t}^{\mathrm{O}}$ denotes the input and output flow rates of TCN node a at time t, respectively; D_{ga}^{C} is a variable representing the connection relationship between line g and node a.

Constraint (50) represents the time when bus i in PDSs is observable. Constraint (51) presents the relationship between the observable time and the observability of PDS buses. Constraints (52)–(54) restrict the transmission direction of the data flow. Constraint (55) restricts the maximum flow rate of each communication line. It is worth noting that if the communication line g connects buses a and b and the direction of data flow is from bus a to bus b, the variables representing the connection relationship would satisfy $D_{ga}^{\mathrm{C}} = 1$ and $D_{gb}^{\mathrm{C}} = -1$.

4) *Availability constraints of SCN:* The transmission capacity and available time constraints for the satellite communication links are given as:

$$1 \le t_i^{\mathrm{S,a}} \le t_i^{\mathrm{S,d}} \le T + 1, \ \ \forall i \in N^{\mathrm{P}} \tag{56}$$

$$t_i^{\mathrm{S,a}} \ge T + 1 - M I_i^{\mathrm{S}}, \ \ \forall i \in N^{\mathrm{P}} \tag{57}$$

$$e_{i,t}^{\mathrm{S}} = \begin{cases} 1, & t_i^{\mathrm{S,a}} \le t \le t_i^{\mathrm{S,d}} \\ 0, \text{otherwise} \end{cases}, \forall i \in N^{\mathrm{P}}, \forall t \tag{58}$$

$$\sum_{i=1}^{n^{\mathrm{P}}} e_{i,t}^{\mathrm{S}} \le M^{\mathrm{S}} C_t^{\mathrm{S}}, \ \ \forall t \tag{59}$$

$$\sum_{i=1}^{n^{\mathrm{P}}} \left(f_i^{\mathrm{C}} e_{i,t}^{\mathrm{S}}\right) \le M^{\mathrm{F}}, \ \ \forall t \tag{60}$$

$$t_i^{\mathrm{S,d}} \ge t_i^{\mathrm{P,C}} \ \ \forall i \in N^{\mathrm{P}} \tag{61}$$

where $t_i^{S,a}$ and $t_i^{S,d}$ are the time periods of constructing and disconnecting satellite communication links for the PDS bus i, respectively; I_i^S is a binary variable and $I_i^S = 1$ indicates that there is a satellite antenna allocated at bus i; M^S is the maximum number of buses using the satellite communication links simultaneously; M^F is the maximum data flow rate; f_i^C is the required data flow rate for bus i.

Constraints (56)–(58) are the conditions of constructing the satellite communication links. Constraints (59) and (60) limit the maximum number of buses using satellite communication links and the data flow rate of those links. Constraint (61) indicates the terrestrial communication links for PDS buses have been repaired before the satellite communication links disconnect. It is worth noting that the satellite communication is not used during the restoration process if $t_i^{S,a} = t_i^{S,d} = T + 1$.

In the above PDS restoration model, the nonlinearized constraints contain (13), (14), (20), (42), (49) and (58). Those constraints can be linearized by the big-M method [30] and then the MILP restoration model for PDSs with STCNs is stated as:

$$\begin{aligned} &\max \sum_{t=1}^{T}\sum_{i\in N^P}\varepsilon_i P_{i,t}^L \tau \\ &\text{s.t. PDS constraints : (7)-(33)} \\ &\quad \text{TCN constraints : (34)-(56)} \\ &\quad \text{SCN constraints : (57)-(62)} \end{aligned} \tag{62}$$

3.3. Solving procedure

The detailed procedure for solving the PDS restoration model is presented in **Algorithm 1**.

Algorithm 1. Solving procedure of PDS restoration model

4. Case study

4.1. Network description

In this paper, the IEEE 33-bus PDS with a STCN is used to verify the efficiency of the proposed model. The area position of the PDS is (120°52′E to 122°12′ E, 30°40′ N to 31°53′ N). The STCN comprises a 12-node TCN and a 1000-satellite LEO SCN to simulate the incomplete coverage characteristic of emerging satellite constellations. As shown in Fig. 8, each node represents a network switch in TCN which transmits the operation information of multiple PDS buses. Furthermore, nine PDS buses are equipped with satellite antennas, which can transmit the information by satellite communication links.

In STCN, the orbit altitude of LEO satellites is 481 km and the minimum elevation angle ε_{min} is 40°. The maximum slant range d_{max} for building the available satellite communication links is set as 714.20 km. The availability rate η_{min} is 75%. The satellite operation simulation is implemented by MATLAB with the Satellite Communication Toolbox [31]. The simulation step size δ is set as 30s and the SGP4 orbit propagator model [32] is utilized in the simulation. For PDS restoration, the time period τ is 10 min and the total restoration time is 200 min. The action time of circuit breakers is 10 min. The time to repair one, two and three terrestrial communication links simultaneously is 20 min, 30 min and 40 min, respectively. The data generated at a PDS bus is 0.5 MB/s. The data flow rate limitations for satellite and terrestrial communication link are 2 MB/s and 10 MB/s respectively. The MILP restoration model is solved by GUROBI. All simulations and model solving are implemented with MATLAB 2022a on a PC with an Intel Core i5–12,400 CPU and 16GB RAM. The constellation simulation is completed within 3 s, and the optimization of the PDS restoration strategy is solved within 10 min each case.

4.2. Simulation results of satellite operation

The available status of satellites for the area of PDSs is analyzed by simulation in Stage 1. We use four terrestrial sites to represent the PDS area, which are Site 1 (120°52′ E, 30°40′ N), Site 2 (122°12′ E, 30°40′ N), Site 3 (120°52′ E, 31°53′ N) and Site 4 (122°12′ E, 31°53′ N). If the four

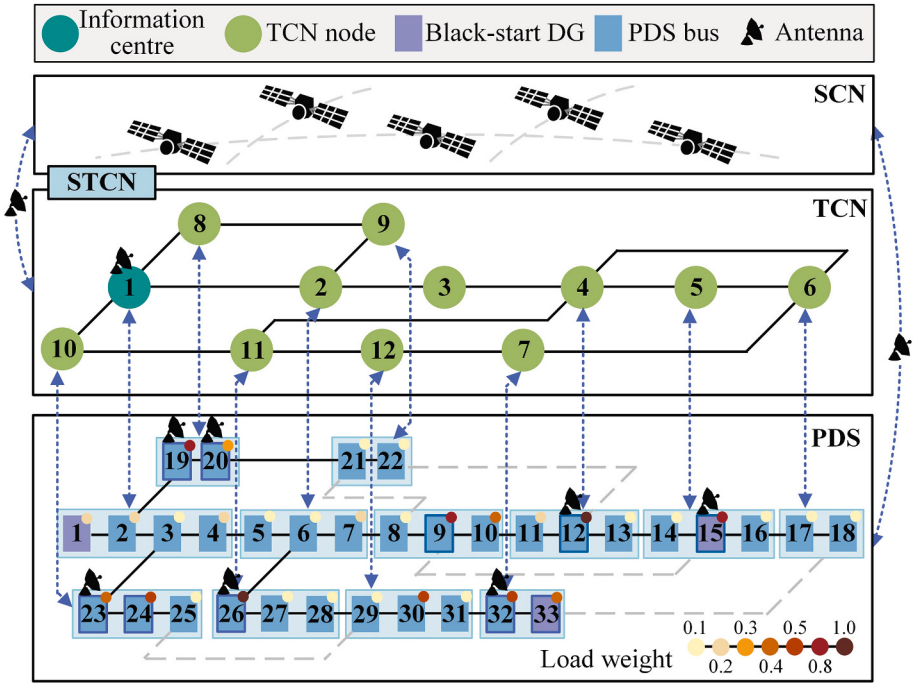


Fig. 8. Topology of the PDS with STCN.

terrestrial sites can build the communication links with the satellites by calculating the satellite connection status, we regard the SCN can be used to restore the PDSs. By satellite operation simulation, three typical scenarios of building satellite communication links are shown in Fig. 9.

In scenario (a), it can be seen that three LEO satellites satisfy the satellite connection conditions for the four sites based on (2). Thus, a total of ten satellite communication links can be built for the PDS area, which are available for PDS restoration in this simulation step. In scenario (b), the distance between each LEO satellite and each site is larger than the maximum required slant range for building satellite communication links. It means that ground communication devices in the PDS area cannot receive the information transmitted by the SCN in this simulation step. In scenario (c), only partial sites can build communication links with LEO satellites. Thus, the SCN is unavailable for PDS restoration in this simulation step.

Each restoration time period in Stage 2 contains 20 satellite simulation steps in Stage 1. Based on the results of satellite connection status, the availability rate of satellite communication links in each restoration time period is given in Table 1. It can be seen that there are 14 time periods in which the availability rate is greater than 75%. It indicates that satellite communication links are available for PDS restoration during those time periods. The satellite connection status and available time periods are regarded as inputs for deriving the optimal restoration strategy in Stage 2.

4.3. PDS restoration results

To verify the effectiveness of the PDS restoration strategy model, the following three cases are designed:

Case 1: The STCN is assumed to provide round-the-clock satellite communication support for the PDS restoration.

Case 2: The STCN provides satellite communication support when the satellite communication links are available for the PDS restoration based on the simulation results in Stage 1.

Case 3: The satellite communication is unavailable.

Then, the PDS restoration model is solved, and the results are shown in Figs. 10 and 11. The restored loads in Cases 1, 2 and 3 are 2.6335MWh, 2.2592MWh, and 2.1295MWh respectively. It indicates that the restoration objective in Case 1 and Case 2 are improved by 23.67% and 6.09% with the assistance of satellite communication.

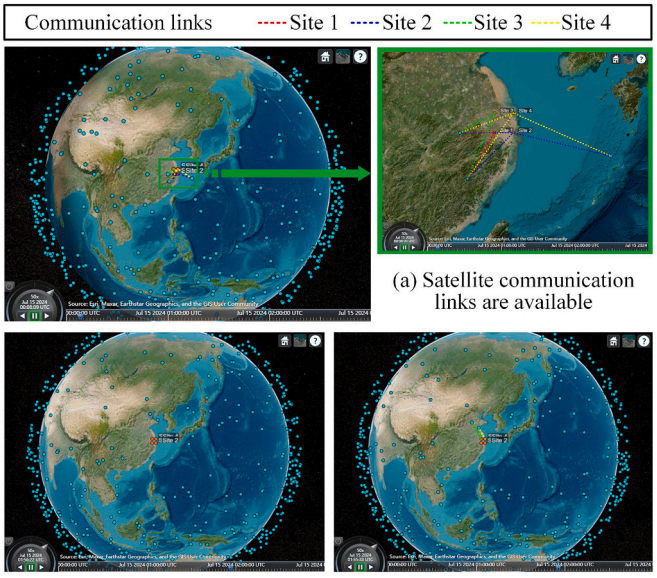


Fig. 9. Typical scenarios for satellite communications.

Table 1
Availability rate of satellite communication.

t	1	2	3	4	5	6	7	8	9	10
η_t^S(%)	80	85	90	80	35	55	80	95	80	55
C_t^S	1	1	1	1	0	0	1	1	1	0
t	11	12	13	14	15	16	17	18	19	20
η_t^S(%)	85	70	80	90	75	80	70	90	90	70
C_t^S	1	0	1	1	1	1	0	1	1	0

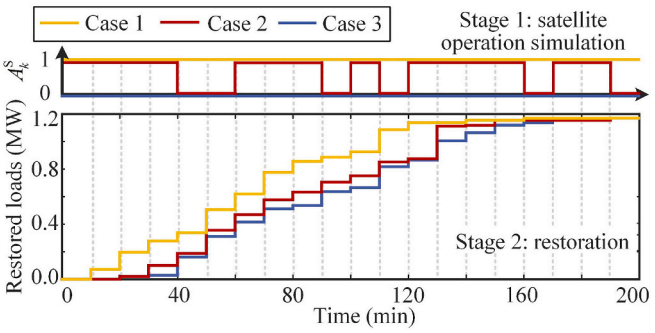


Fig. 10. Restoration curves in the three cases.

Furthermore, the restoration efficacy in Cases 1 and 2 is higher than that in Case 3 by comparing the restoration curves in Fig. 10. In particular, the restoration process in Case 1 is completed 30 min earlier than that in Cases 2 and 3. It indicates that the efficacy of PDS restoration is improved by introducing satellite communications.

In Fig. 11, the restoration sequences of buses, power lines and terrestrial communication links in the PDS with STCN are presented. It can be seen that building available satellite communication links affects optimal restoration sequences. For example, in Case 2, the loads at buses 19, 20 and 26 in PDSs are restored by introducing satellite communication. The restoration time of those buses are delayed at least 10 min in Case 3 because the recovery of observability relying on the TCN needs more time. Furthermore, there are some waiting time periods in restoration, such as 120 min–130 min and 140 min–150 min in Case 1, 150 min–170 min in Case 2, and 170 min–180 min in Case 3. The reason is that the buses are unobservable in PDSs due to the damaged terrestrial communication links although the power lines have been restored. Thus, the maintenance crew is restoring those communication links in TCN in the waiting time period.

Meanwhile, the restoration network topologies in the three cases are shown in Fig. 12. It can be seen that the topologies of restored microgrids are different in the three cases. In Case 1, the microgrid centred by the black-start DGs of buses 1, 15 and 33 contain 19, 10 and 4 buses, respectively. In comparison, in Case 2 and Case 3, the microgrid centred by bus 1 is larger which contains 27 buses. Other microgrids centred by buses 15 and 33 are smaller, which contain 4 buses and 2 buses, respectively. It indicates that the satellite communication influences the structure of restored power networks.

Furthermore, the restored topologies of terrestrial TCN are also affected by the satellite communication in those cases. In Case 1, nodes 7 and 8 in TCN are not contained in the restored topology because the information transmission through those nodes is performed by the satellite communication links in SCN. In Cases 2 and 3, the connection relationship of some terrestrial communication links is different in the restored TCN, which is affected by the access of satellite communication.

4.4. Influences of satellite operations

1. Influences of satellite connection status

The rotation of the Earth and the fast movement of satellites make

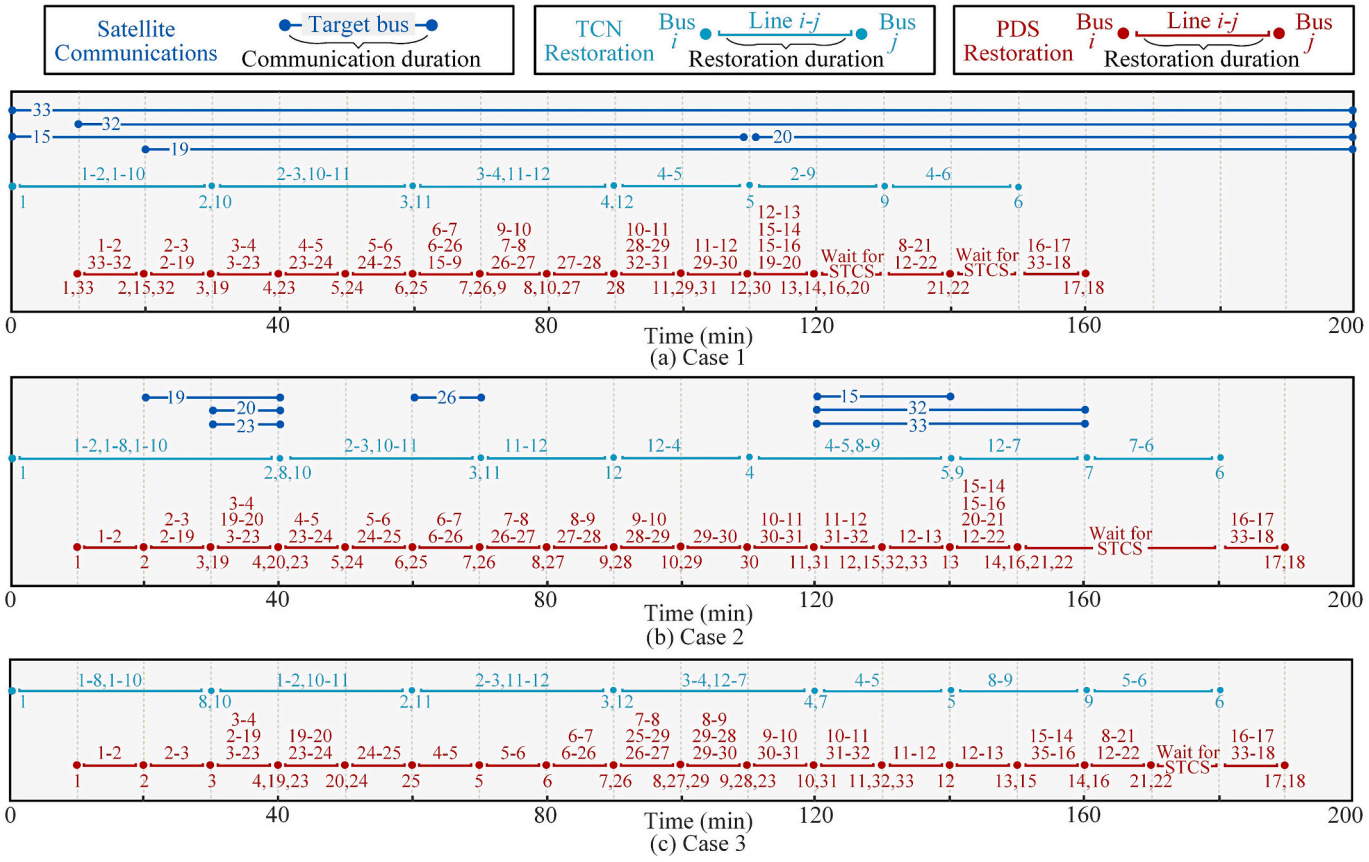


Fig. 11. Restoration sequences of buses, power lines and terrestrial communication links in the three cases.

the relative positions between LEO satellites and ground devices constantly changed. Thus, the satellite connection status is varied with time. In this part, we further analyze the influences of the satellite connection status by changing the restoration start time. The satellite connection status during 24 h is derived in Stage 1 and then the restoration results are derived when each hour is regarded as the start time for restoration in Stage 2. It is worth noting that the restoration start time with less than 200 min left in 24 h is not considered in this case. The restored load results are presented in Fig. 13.

It can be seen that different satellite connection statuses caused by the restoration start time directly affect the restoration results. The restored loads in PDSs range from 128.34MWmin at 3:00 to 135.55MWmin at 15:00 with start times varying from 0:00 to 20:00. Furthermore, restored loads have a positive correlation with the availability rate of satellite communication during PDS restoration. The availability rate of satellite communication is low from 3:00 to 7:00, so the restored loads are smaller when the restoration start time varies from 3:00 to 5:00 compared with those in other restoration start times (e.g., from 6:00 to 9:00).

2. Influences of the minimum elevation angle

In this part, we analyze the influences of the minimum elevation angle on the restoration results. We increase the minimum elevation angle from 20° to 50° and the variation of the maximum slant range d_{max} is shown in Table 2. It can be seen that the maximum slant range is reduced with increasing minimum elevation angle. Then, we set the start time of the restoration as 15:00 and present the availability rate of satellite communication in Fig. 14 and restoration results in Fig. 15 by changing the minimum elevation angle.

The availability rate for satellite communication decreases as the minimum elevation angle increases. When the minimum elevation angle is 20°, the satellite communication links are always available during the restoration. As the minimum elevation angle increases, the satellite communication links gradually become unavailable in more and more time periods. Meanwhile, as the minimum elevation angle increases from 20° to 50°, the restored loads are smoothly decreased from 2.6335MWh to 2.1295MWh. It indicates that the minimum elevation angle directly affects the availability rate of satellite communication and then the restoration effect are influenced. It implies that the LEO satellite constellation and parameters have significant influences on the interaction between PDSs and STCNs, which should be concerned in the application of satellite communication in PDS restoration.

5. Conclusion

This paper proposes a two-stage restoration framework for PDSs with integrated STCNs. In Stage 1, a simulation model for LEO satellites is established to derive the available satellite communication links. Then, a MILP model is established to obtain the optimal restoration strategy for PDSs in Stage 2. The operation constraints of the PDSs and STCNs are all considered in the proposed model. The results demonstrate that:

1) The effectiveness of the proposed restoration framework is verified under the cyber-physical interaction process in PDSs with STCNs. The available satellite communication links derived in Stage 1 provide more flexibility for designing the optimal restoration strategy of PDSs in Stage 2.
2) By the optimal restoration strategy, the restored loads increase by 23.67% with the assistance of round-the-clock satellite communication. It indicates that the satellite communication is an effective measure supporting emergency communication of power systems in various extreme situations.
3) The available statuses and minimum elevation angle of the LEO satellites have significant influences on the restored loads of PDSs,

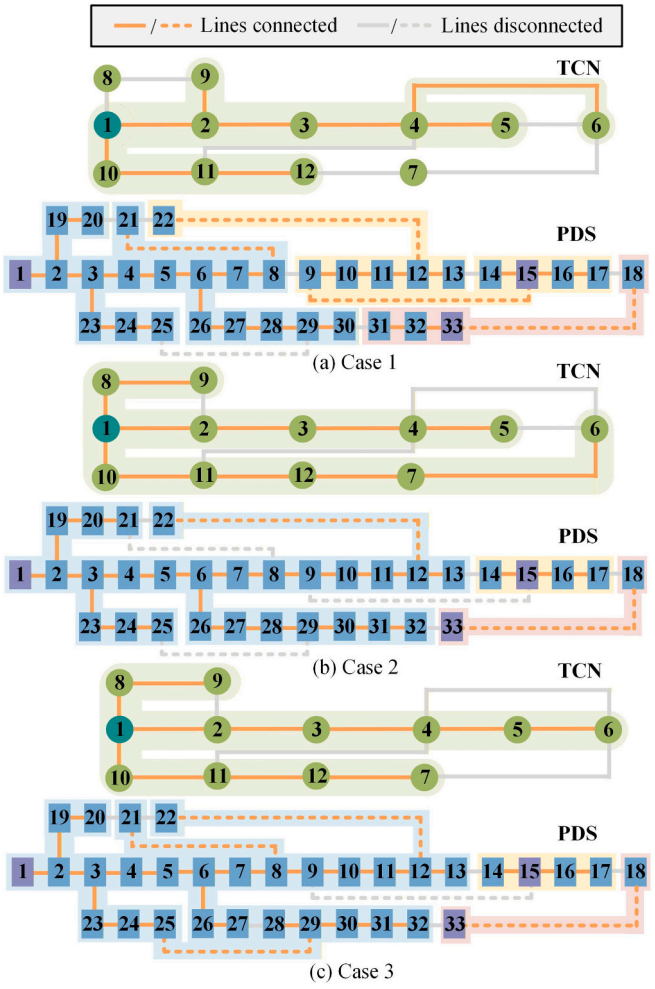


Fig. 12. Restoration topologies in the three cases.

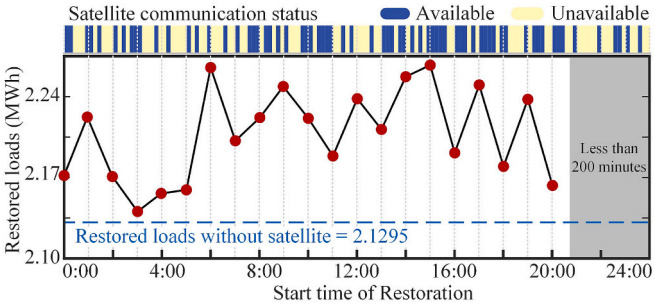


Fig. 13. Restoration results at various start times.

Table 2
Relationship between slant range and elevation angle.

ε_{min} (°)	20	25	30	35	40	45	50
d_{max} (km)	1153.5	996.4	877.3	785.7	714.2	657.8	613.1

which need more attention in the design of a LEO-based satellite constellation for power grid operation.

Our future work will focus on the application of LEO-based satellite communication to improve power system resilience and further consider the impacts of various uncertainties [33] in system operation.

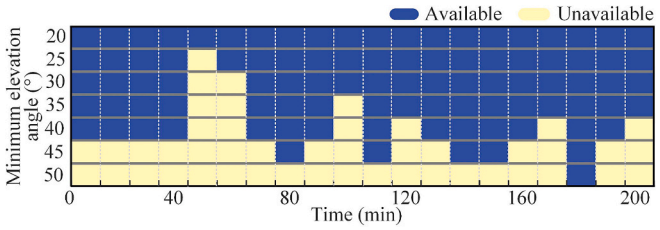


Fig. 14. Satellite operation results with different minimum elevation angles.

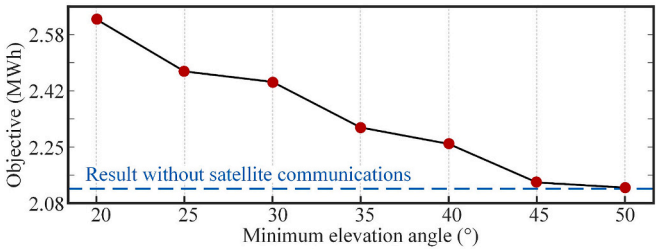


Fig. 15. Restoration results with different minimum elevation angles.

CRediT authorship contribution statement

Han Wang: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Zhe Hu:** Visualization, Validation, Formal analysis, Data curation. **Yunpeng Xiao:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation. **Quan Zhou:** Writing – review & editing, Validation, Supervision. **Hongxun Hui:** Visualization, Validation, Methodology, Conceptualization. **Zheng Yan:** Validation, Supervision, Methodology. **Zhiguo Huang:** Visualization, Methodology, Investigation. **Jize Han:** Visualization, Validation, Supervision, Conceptualization. **Xu Jin:** Visualization, Software, Resources, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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